

Sri Lanka Institute of Information Technology



Human Fitness Prediction Artificial Intelligence and Machine Learning| IT2011

Year 2 semester 1 (2025)

Group Members

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Document Contents

1. Introduction and Problem Statement	3
2. Dataset Description.....	3
3. Pre-processing and EDA	4
a) Initial Data Cleaning and Type Conversion.....	4
b) Encoding	5
c) Outlier Removal.....	6
d) Feature Engineering.....	7
e) Feature Scaling	8
f) Feature Selection	9
4. Model Design and Implementation	11
5. Evaluation and Comparison	11
6. Ethical Considerations and Bias Mitigation	12
6.2 Mitigation Strategies	14
7. Reflections and Lessons Learned	14
8. References.....	15
 6.1 Potential Ethical Risks	14

1. Introduction and Problem Statement

In recent years, the integration of data-driven methodologies into the field of health and fitness has gained significant importance. Assessing an individual's fitness level is a multifaceted process that depends on a variety of physiological, behavioral, and lifestyle factors. Traditional assessment methods often rely on subjective evaluations or manual data analysis, which can lead to inconsistencies and inefficiencies. Consequently, the application of advanced computational techniques, particularly Machine Learning (ML), offers a promising solution to enhance the accuracy and reliability of fitness evaluations.

The primary aim of this study is to develop a Machine Learning-based classification model capable of accurately predicting whether an individual is fit or not fit based on various health and lifestyle attributes. This model seeks to enhance the reliability of fitness evaluations, demonstrate the effectiveness of ML techniques in handling real-world data complexities, and contribute to the advancement of data-driven approaches in personal health and wellness analysis.

2. Dataset Description

The dataset used in this study is the Real-World Style Fitness Classification Dataset, a synthetic dataset designed to simulate realistic health and lifestyle information. It represents a binary classification problem in which the objective is to predict whether a person is fit ($\text{is_fit} = 1$) or not fit ($\text{is_fit} = 0$) based on various physiological and behavioral attributes such as activity level, sleep duration, calorie intake, and other health-related features.

This dataset closely mimics real-world health data, making it suitable for developing and evaluating machine learning models aimed at fitness prediction and health analytics. It is publicly available and serves as an excellent resource for educational and research purposes, particularly for experiments involving data preprocessing, feature engineering, model selection, and performance evaluation in classification problems.

a) Dataset Overview

- Simulated real-world binary classification dataset for predicting individual fitness levels.
- Contains approximately **2,000 samples** with both numerical and categorical features.
- Target variable: **is_fit**, indicating whether a person is fit (1) or not fit (0).
- Includes **10 features** reflecting various health and lifestyle aspects such as age, height, weight, blood pressure, and activity level.
- Data contains **mixed data types, missing values, outliers, and noisy entries** to emulate real-world data imperfections.

- Structured in a **tabular format**, suitable for traditional machine learning and neural network models.

b) Dataset Justification

- The dataset offers a **realistic simulation of health data challenges**, including inconsistencies, non-linear feature relationships, and complex interactions.
- Ideal for **binary classification tasks** focused on predicting health or fitness outcomes.
- Enables practical application of **data cleaning, encoding, scaling, and outlier detection** techniques.
- Well-suited for developing **robust machine learning models** due to its moderate size and diversity of features.
- Provides an opportunity to evaluate model performance under real-world-like data noise and imbalance conditions.
- The dataset's design encourages experimentation and comparison of multiple classification algorithms.

3. Pre-processing and EDA

a) Initial data cleaning and type conversion

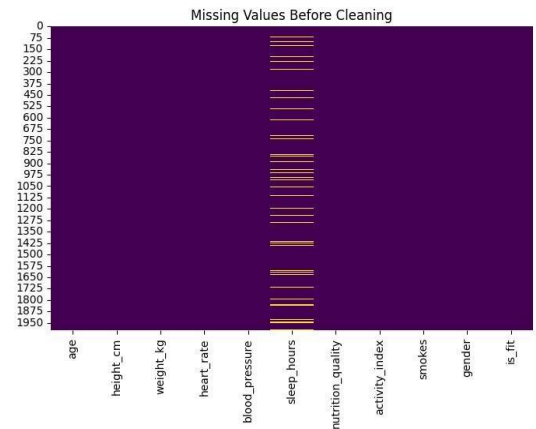
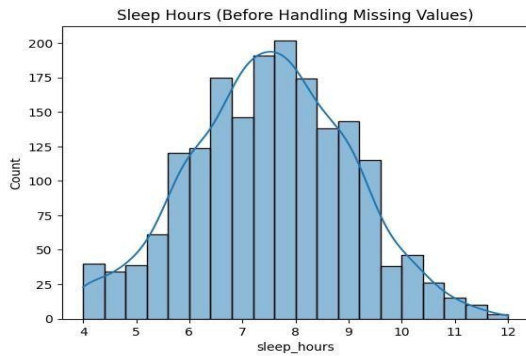
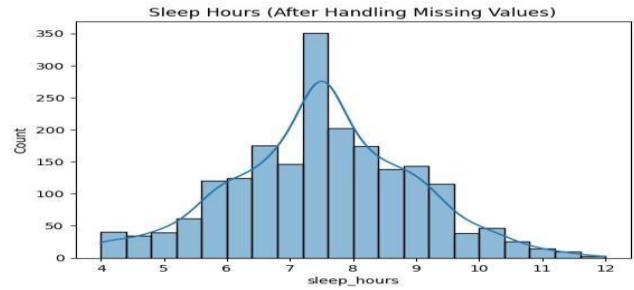
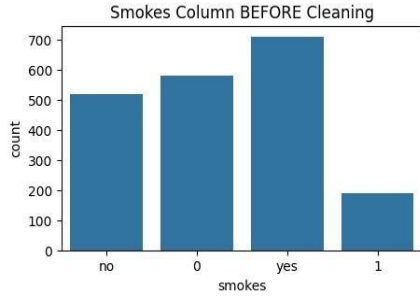
Handling mixed data types in smokes column:

The smokes column contained inconsistent entries (0, 1, "yes", "no"). To ensure uniformity, all values were first standardized to consistent string labels ("no" for non-smokers, "yes" for smokers), then converted to a uniform binary format during encoding.

Addressing missing values in sleep_hours:

Approximately 8% of the sleep_hours entries were missing (NaN). These were imputed using the median of the column to preserve the central tendency while minimizing distortion from outliers.

b)



Encoding

Categorical variables were encoded using **Label Encoding** to convert them into numerical form suitable for machine learning models.

- **Gender Encoding:**

- 'F' → 0 ◦

- 'M' → 1

This binary mapping maintains a compact representation without increasing dimensionality, which is ideal for smaller datasets and neural network architectures.

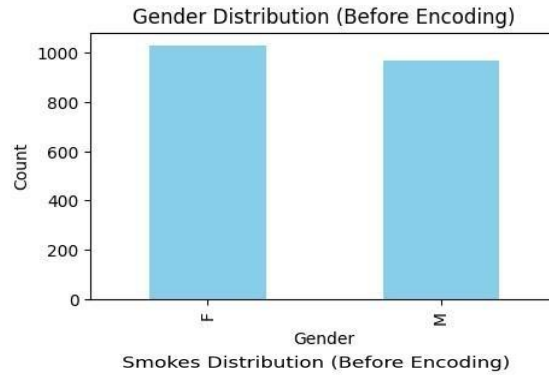
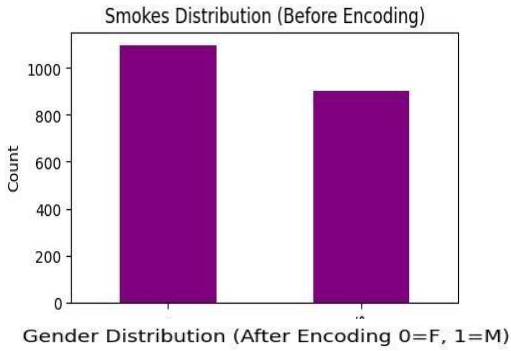
- **Smokes Encoding:**

After standardization, all entries were encoded as:

- "no" → 0 ◦

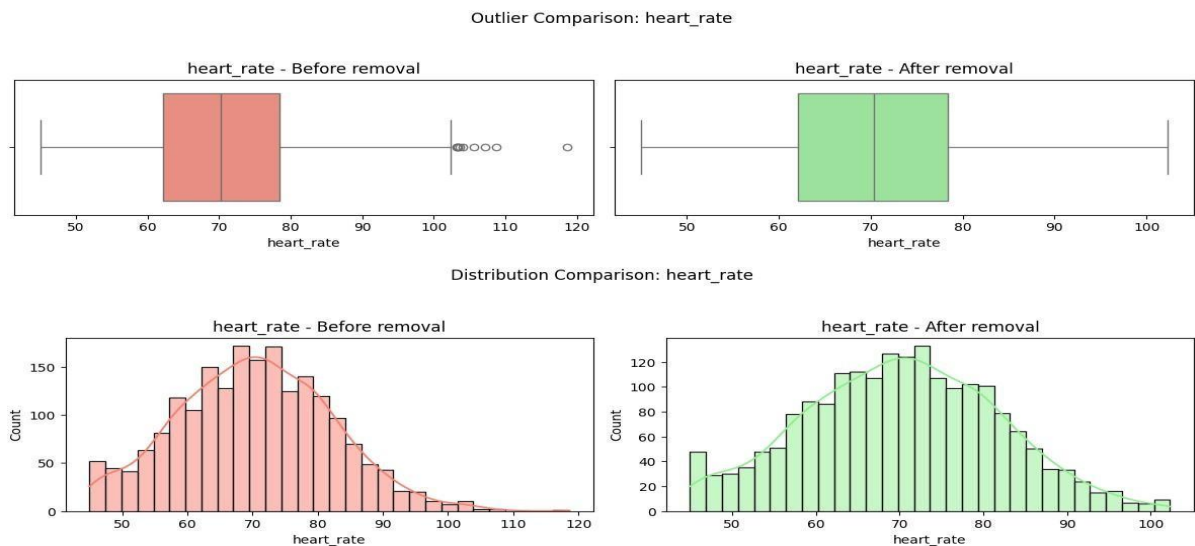
- "yes" → 1

This transformation ensured numerical consistency across all features while preserving semantic meaning for downstream modeling processes.



c) Outlier Removal

The **weight_kg** column contained a small number of biologically implausible outliers (values below 30 kg or above 200 kg). These were identified and removed using the **Interquartile Range (IQR)** method. Approximately 2% of the samples were excluded during this process, leading to more realistic BMI calculations. Other numerical features, such as **heart_rate** and **blood_pressure**, were inspected for outliers but retained, as their distributions remained within medically acceptable ranges.

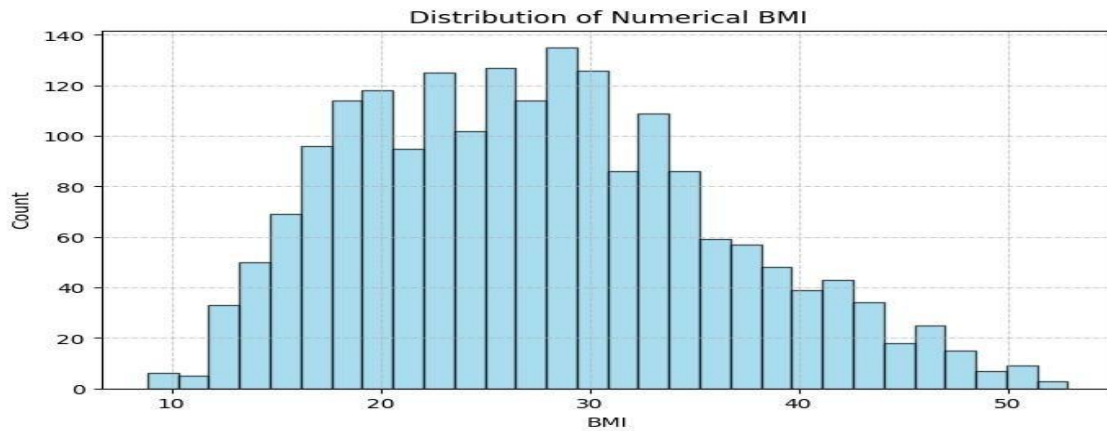


d) Feature Engineering

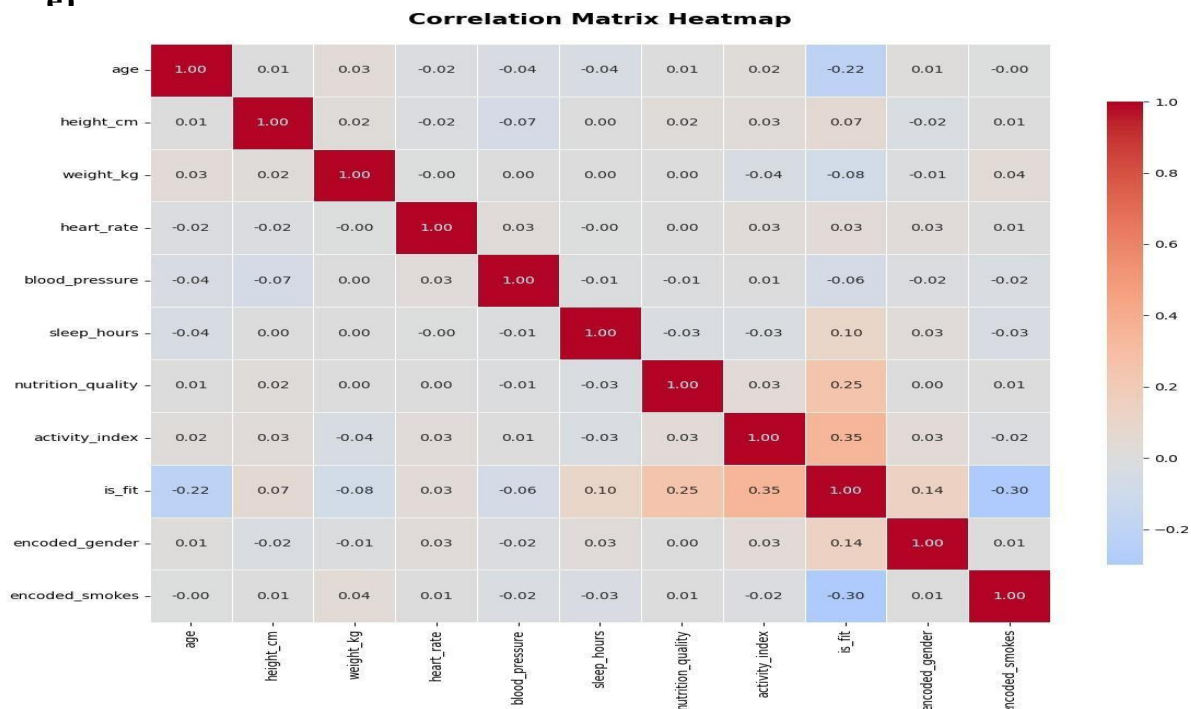
A new derived feature, **Body Mass Index (BMI)**, was computed using the formula:

$$\text{BMI} = \frac{\text{weight_kg}}{(\text{height_cm}/100)^2}$$

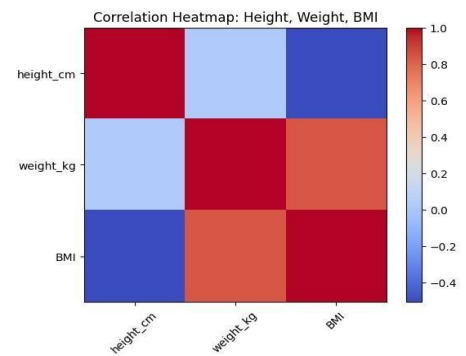
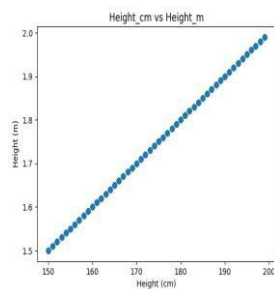
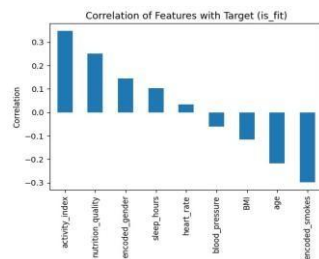
BMI serves as a more interpretable indicator of body composition and is a strong predictor of overall fitness level. Behavioral and lifestyle-related features, such as **nutrition_quality** and **activity_index**, were retained in their original form as they already represent normalized and aggregated measures. Preliminary statistical analysis indicated that features such as **BMI**, **activity_index**, and **smokes** show strong differentiation between the fitness classes ($\text{is_fit} = 0$ and $\text{is_fit} = 1$), justifying their inclusion in the model.

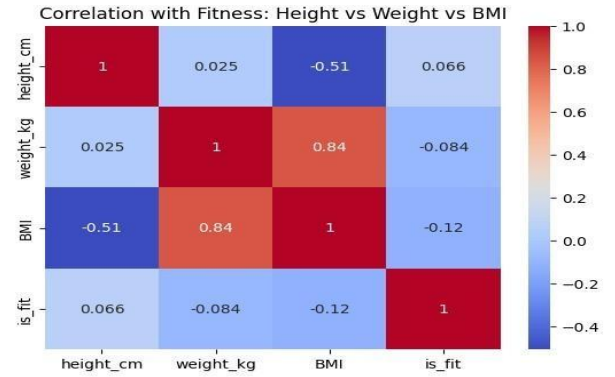
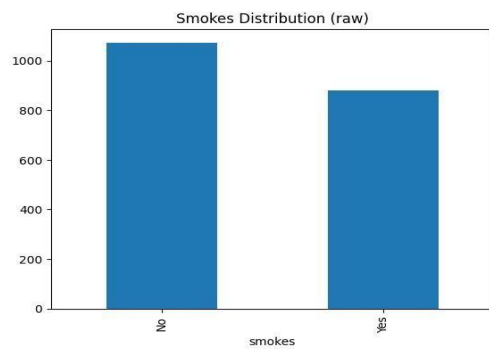


a)



e) Feature Selection

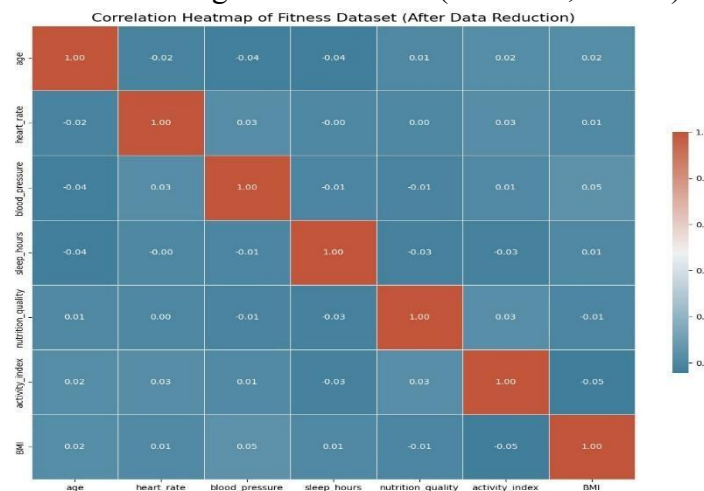


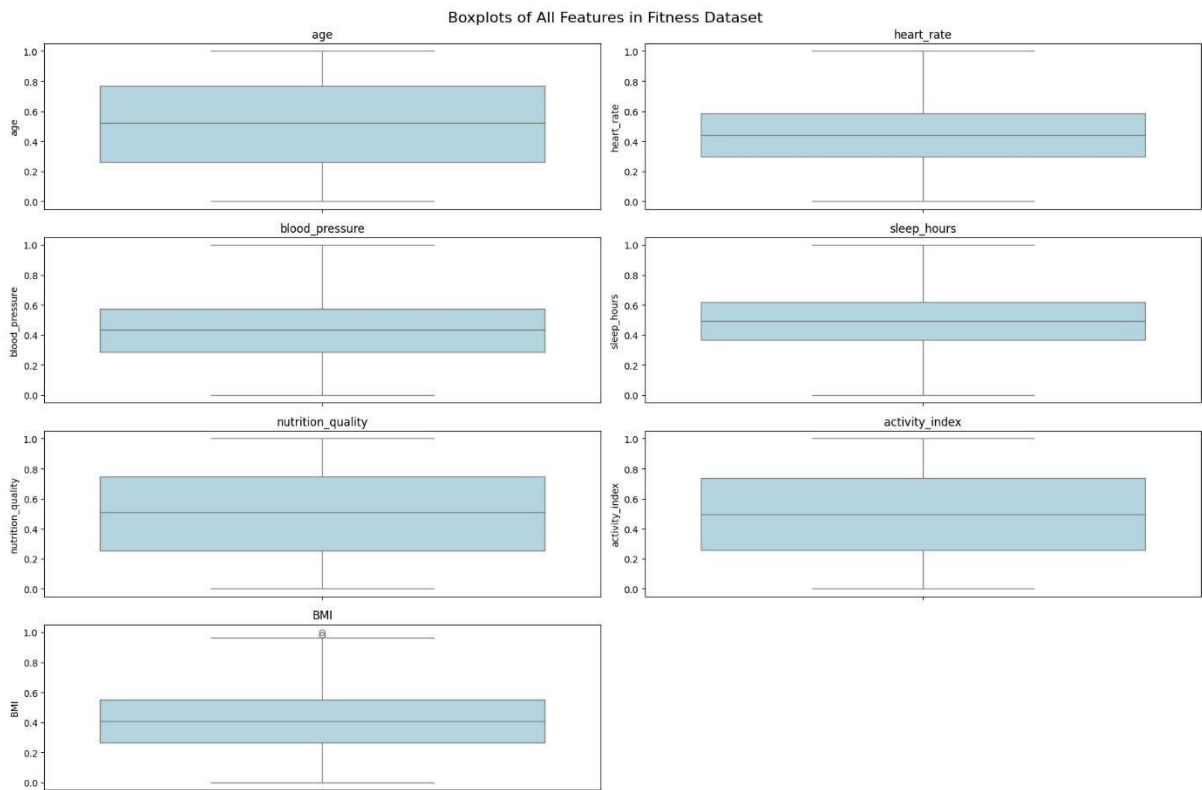
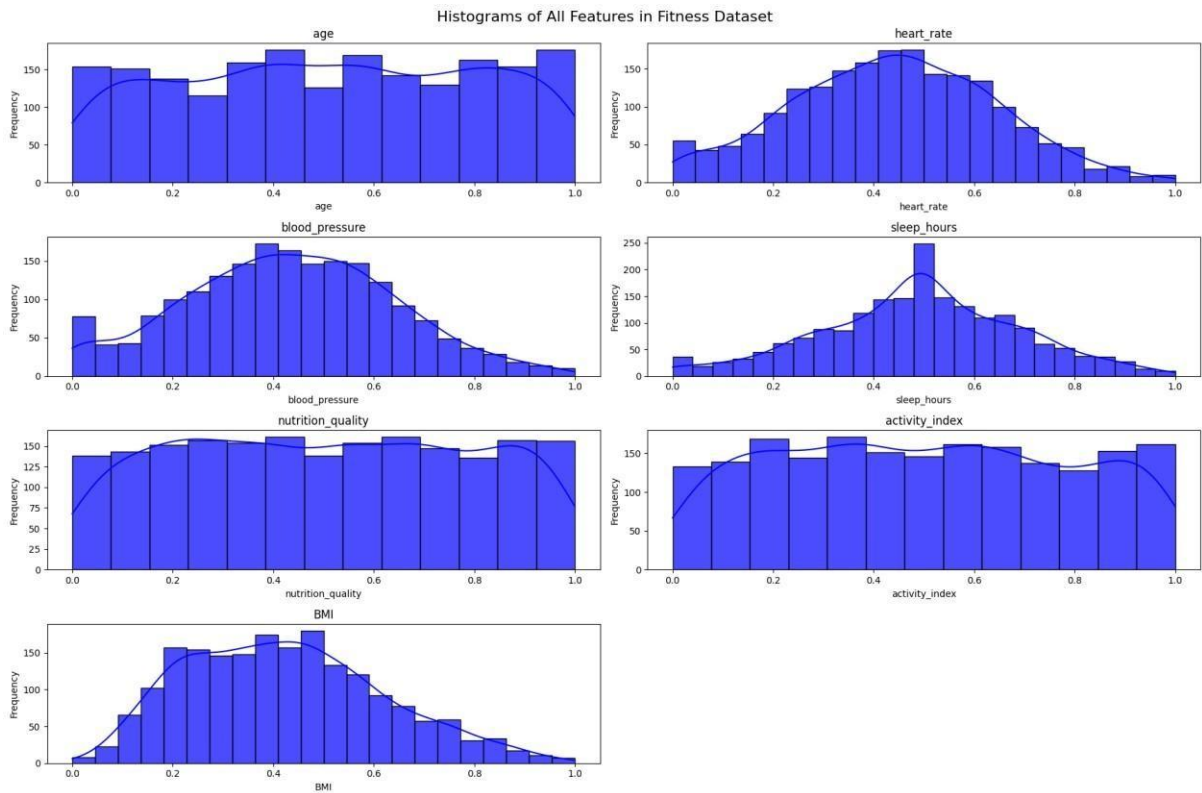


f) Feature Scaling

Standardization of numerical features:

To ensure equal influence during model training—especially for distance-based or gradientbased algorithms (e.g., neural networks, SVMs)—all numerical features (age, height_cm, weight_kg, BMI, heart_rate, blood_pressure, sleep_hours, nutrition_quality, activity_index) were scaled using StandardScaler (mean = 0, std = 1).





4. Model Design and Implementation

The prediction task is a binary classification problem. Several well-established machine learning models were implemented and evaluated using a standard 80/20 train-test split (test size=0.2).

Models Implemented:

1. Random Forest
2. Decision Tree
3. Logistic Regression
4. Support Vector Machine (SVM)
5. Naive bayes
6. K-Nearest Neighbors (KNN)

5.valuation and Comparison

Model	Test Accuracy (%)	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.7545	0.6974	0.6795	0.6883	0.7419
Decision Tree	0.7289	0.6543	0.6794	0.6666	0.7577
Logistic Regression	0.8030	0.8139	0.6645	0.7317	0.7807
SVM	0.7672	0.6923	0.7500	0.7200	0.8486
naive bayes	0.7900	0.7890	0.6470	0.7111	0.7666

K-Nearest Neighbors (KNN)	0.7596	0.7460	0.6026	0.6667	0.8202
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Comparison

Among all tested models, Logistic Regression achieved the best overall performance with a test accuracy of 80.3%, precision of 0.81, recall of 0.66, and an F1-score of 0.73.

While models such as SVM and Random Forest showed competitive results, Logistic Regression demonstrated a superior balance between precision and recall with the highest F1-score.

Additionally, it is computationally efficient and offers better interpretability, allowing clear insight into how each feature contributes to predictions.

Therefore, Logistic Regression was selected as the final model.

Selected Model- Logistic Regression;

6. Ethical Considerations and Bias Mitigation

6.1 Potential Ethical Risks

Although the dataset used in this project is synthetic and does not contain any real personal health information, several ethical issues may still arise during model development, interpretation, and communication of results. These potential risks are outlined below.

(a) Bias in Feature Representation

Certain attributes such as *gender* and *smoking status* can introduce unintentional bias. For instance, if the dataset distribution favors one group (e.g., males or non-smokers), the model may learn patterns that disproportionately classify certain categories as more or less fit, reflecting biased predictions.

(b) Misinterpretation of Results

Since this dataset imitates health-related variables, there is a risk that the results may be misinterpreted as medically valid. The predictive outcomes of the model are educational in nature and should not be considered diagnostic or clinically accurate.

(c) Privacy and Data Protection

Even though the dataset is synthetic, the project highlights ethical considerations that apply to real-world health data, such as the need for informed consent, anonymization, and

compliance with data protection regulations (e.g., GDPR or HIPAA) when using sensitive personal information.

6.2 Mitigation Strategies

To address the potential ethical and bias-related risks, the following strategies were implemented or are recommended for similar real-world applications.

(a) Fair Data Preprocessing

The *smokes* column, which contained mixed data types (numeric and string), was standardized to a consistent binary format to prevent misclassification. Similarly, categorical features such as *gender* were label-encoded in a non-hierarchical manner to avoid ordinal bias.

(b) Handling Class Imbalance

Performance evaluation employed balanced metrics including **F1-score**, **precision**, **recall**, and **ROC-AUC**, rather than relying solely on accuracy. Cross-validation and stratified sampling techniques were used to maintain proportional class representation during training.

(c) Transparent Communication

All model findings are clearly documented as educational outputs. A disclaimer was provided emphasizing that the dataset is entirely synthetic and unsuitable for real-world medical prediction or diagnostic use.

(d) Ethical Model Evaluation

Feature importance and confusion matrices were examined to ensure that predictions were not disproportionately influenced by sensitive variables. Future work involving real data should include fairness audits and explainability techniques (e.g., SHAP or LIME) to ensure ethical transparency.

(e) Responsible Use and Awareness

Students and practitioners are encouraged to understand that technical accuracy does not automatically imply fairness or reliability. Continuous awareness of ethical and social impacts is essential in applying machine learning to health-related data.

7. Reflections and Lessons Learned

7.1 Data Challenges and Preprocessing

This project provided a realistic understanding of how messy data can affect model performance. Handling mixed data types, imputing missing *sleep_hours* values, and detecting outliers in *weight_kg* were crucial in achieving stable and reliable models. The process reinforced the importance of thorough preprocessing before model training.

7.2 Importance of Ethical Awareness

The project emphasized that ethical considerations are integral to data science practice. Even with synthetic data, understanding the implications of bias, fairness, and interpretability is vital to ensure responsible AI development.

7.3 Handling Imbalanced Data

Dealing with a 60:40 class imbalance demonstrated the need to evaluate model performance beyond accuracy. Metrics such as F1-score and ROC-AUC provided a more comprehensive measure of model reliability and fairness.

7.4 Feature Engineering and Model Improvement

Feature engineering such as computing derived variables like BMI and scaling continuous features significantly enhanced model interpretability and generalization. These steps simulated realistic data science workflows.

7.5 Broader Learning Experience

Through this dataset, the project offered an end-to-end understanding of the machine learning pipeline from raw data cleaning and feature encoding to ethical reflection and evaluation. It highlighted that successful ML models depend not only on computational performance but also on their societal and ethical impact.

7.6 Key Takeaway

The most important lesson learned is that **ethical awareness, fairness, and transparency** must accompany technical excellence in data science. Building responsible models ensures that machine learning contributes positively to real-world decision-making, even when working with synthetic or educational datasets

8. References

- **Kaggle Fitness Classification Dataset**

<https://www.kaggle.com/datasets/muhammedderric/fitness-classification-datasetsynthetic/suggestions>

- scikit-learn developers. (2024). Scikit-learn User Guide. Available at: • https://scikit-learn.org/stable/user_guide.html